

DD: SUB-WF-SUSPEND-NP-01 — Subscription Non-Payment Suspension

Developer implementation guide:

DD_SUB-WF-SUSPEND-NP-01-Subscription_Non_Payment_Suspension-DEV_IMPL_GUIDE-v1.0.md.

1. Purpose

This rewritten DD is the developer-facing version of the module contract

DD_SUB-WF-SUSPEND-NP-01-Subscription_Non_Payment_Suspension-v1.0.md.

Verdict: ■ New | **Wave:** 1 | **Group I:** Subscription Management | **Version:** v1.0 Operation that transitions an ACTIVE subscription to SUSPENDED due to non-payment, on instruction from BIL-04 (the dunning engine). Owns: the dunning-only trigger gating, the `originIntent = SUSPEND_NP` marker that protects the suspension from voluntary resume, the per-operator config catalog, the trigger REST endpoint, the rich domain Kafka events. Composes the framework primitives from **DD_SUB-WF-FRAMEWORK-01 + DD_SUB-WF-FRAMEWORK-01-WORKERS**. **Distinct from SUB-WF-PAUSE-01:** SUB-WF-PAUSE-01 is voluntary (customer/admin-initiated) pause with a fee and customer-resumable. SUB-WF-SUSPEND-NP-01 is system-driven from dunning, no fee, resumable only via SUB-WF-RESUME-01's DUNNING_RESUME trigger (after debt settlement) or ADMIN_FORCE_RESUME (exceptional admin override). Both flows transition the subscription to the same SUSPENDED status — the distinction is preserved in the pause-history row's `originIntent` field. **Companion DDs** (per-operator BPMN satellites): - **DD_SUB-WF-SUSPEND-NP-01-FLOW-WIK** — KE WIK reference BPMN (shipped v1.0) - **DD_SUB-WF-SUSPEND-NP-01-FLOW-WUG / -WTZ / -YASSN** — pending operator deep dives

The goal of this rewrite is to make the DD easier for a mid-level developer to implement without losing the detailed source contract.

2. Implementation Scope

This DD should be implemented as part of the owning SOPHIX capability, not as an isolated service unless the deployment architecture explicitly separates that capability.

Implement this DD with these rules:

- Use the owning module APIs/repositories for authoritative writes.
- Use approved internal APIs/client wrappers when reading another module's data.
- Do not query another module's database tables directly from business flow code.
- Treat worker names as logical Camunda external-task topics, not separate deployments.
- One worker application may handle multiple topics, but metrics, retries, and logs must remain visible per topic.
- Emit success events only after the authoritative state commit succeeds.
- Use outbox publishing for Kafka events.

3. Runtime Metadata

Item	Value
Source file	/Users/macbook/Downloads/Sophix_V3_BSS_DDs_MD_2026-05-27/04_subscription_workflows/DD_SUB-WF-SUSPEND-NP-01-Subscription_Non_Payment_Suspension-v1.0.md
Version	v1.0
DD type	module contract
BPMN/process keys	Not explicitly defined
Drools packages	rules.subscription.suspend-np.<operator>

4. Runtime Contracts To Implement

4.1 APIs

- POST /api/subscriptions/{subscription_id}/suspend-np

4.2 Tables

- subscription_suspend_np_config

For each table in this DD, implement the schema with:

- a short table comment explaining what the table is for;
- column comments or migration notes explaining each field;
- constraints and indexes from the DD;
- seed data where the DD provides operator-specific sample rows.

4.3 Worker Topics

- bil01-emit-intent
- drools-decide
- external-cancel-requested
- ful-call-restrict
- kafka-emit
- notification-send
- sub-lm-cancel-in-flight-operation
- sub-lm-commit-state
- sub-lm-compute-cycle-window
- sub-lm-put-active-restrictions
- sub-lm-put-master-fields
- sub-lm-put-pending-status
- subscription-operation-finalize

Worker-topic guidance:

- A BPMN service task uses the topic.
- A worker application subscribes to the topic.
- The topic handler validates input variables, calls the owning module/API, writes outputs back to Camunda, and records metrics.
- Do not treat the topic string as a shell command or Kubernetes deployment name.

4.4 Events

- SubscriptionSuspendNPRejected
- SubscriptionSuspendedForNonPayment

Event guidance:

- Write events through the transactional outbox.
- Include operation id, aggregate id, operator code, actor, and correlation data.
- Consumers should be able to rebuild downstream state without querying workflow internals.

5. Developer Implementation Checklist

1. Add or verify database migrations and seed rows.
2. Add or verify config rows for the operator/module.
3. Implement request/command DTOs and validation.
4. Implement idempotency and in-flight operation checks where the DD requires them.

5. Implement Drools fact mapping and package deployment where rules are used.
6. Implement BPMN process, service tasks, gateways, timers, and message catches where the DD is a flow.
7. Implement worker-topic handlers using owning module APIs.
8. Implement compensation paths and manual recovery paths.
9. Emit success/rejected events through the outbox.
10. Add smoke tests for happy path, validation failure, dependency failure, and retry/idempotency.

6. Infrastructure And AWS Notes

Deploy this DD with the existing SOPHIX platform components unless the owning module requires isolation:

Concern	Recommendation
API/runtime	Run inside the owning module deployment or existing workflow API pods.
Workers	Consolidate low-volume topics into shared worker apps; keep per-topic metrics.
Database	Use the owning module PostgreSQL schema and migration pipeline.
Kafka	Use the foundation Kafka/MSK setup and transactional outbox.
Camunda	Deploy BPMN files through the workflow deployment pipeline.
Drools	Deploy KJAR/rule artifacts through the approved rules pipeline.
Secrets	Use the platform secret manager; on AWS prefer Secrets Manager/Parameter Store with IRSA.
Observability	Alert on stuck operations, failed workers, outbox backlog, and external dependency failures.

7. Original DD Detail Preserved

The following section preserves the detailed source contract for implementation and review.

Verdict: ■ New | **Wave:** 1 | **Group I:** Subscription Management | **Version:** v1.0

Operation that transitions an ACTIVE subscription to SUSPENDED due to non-payment, on instruction from BIL-04 (the dunning engine). Owns: the dunning-only trigger gating, the `originIntent = SUSPEND_NP` marker that protects the suspension from voluntary resume, the per-operator config catalog, the trigger REST endpoint, the rich domain Kafka events. Composes the framework primitives from **DD_SUB-WF-FRAMEWORK-01 + DD_SUB-WF-FRAMEWORK-01-WORKERS**.

Distinct from SUB-WF-PAUSE-01: SUB-WF-PAUSE-01 is voluntary (customer-/admin-initiated) pause with a fee and customer-resumable. SUB-WF-SUSPEND-NP-01 is system-driven from dunning, no fee, resumable only via SUB-WF-RESUME-01's DUNNING_RESUME trigger (after debt settlement) or ADMIN_FORCE_RESUME (exceptional admin override). Both flows transition the subscription to the same SUSPENDED status — the distinction is preserved in the pause-history row's `originIntent` field.

Companion DDs (per-operator BPMN satellites):

- **DD_SUB-WF-SUSPEND-NP-01-FLOW-WIK** — KE WIK reference BPMN (shipped v1.0)
- **DD_SUB-WF-SUSPEND-NP-01-FLOW-WUG / -WTZ / -YASSN** — pending operator deep dives

Glossary

Term	Meaning
Non-payment suspension	Full revocation of service for failure to pay, driven by BIL-04 dunning at the configured escalation level (typically L3 in a 3-tier ladder). Distinct from RESTRICT (partial constraint, ACTIVE status preserved) and from TERMINATE (permanent, terminal status). Equipment stays in place; no retrieval WO is dispatched.
<code>originIntent = SUSPEND_NP</code>	Marker on the <code>subscription_pause_history</code> row created by this workflow. Distinguishes a non-payment suspension from a voluntary pause (<code>originIntent = CUSTOMER_REQUESTED_PAUSE</code> or <code>ADMIN_PAUSE</code>). Drives downstream rules: SUB-WF-RESUME-01 only accepts DUNNING_RESUME or ADMIN_FORCE_RESUME triggers on SUSPEND_NP-flagged suspensions.

Term	Meaning
<code>systemManaged = true</code>	Companion flag on the pause-history row indicating the suspension was applied by automated dunning. Prevents human-actor reactivation through voluntary resume paths.
Dunning reason code	Free-text reason supplied by BIL-04 (e.g., <code>DUNNING_GRACE_EXPIRED</code> , <code>DUNNING_WALLET_DEPLETED</code> , <code>DUNNING_ESCALATION_T3</code>). The framework does not enumerate; BIL-04's catalog defines them. Propagated to <code>subscription.current_transition_reason_code</code> and the Kafka event.
Dunning escalation level	Integer level in the operator's tiered dunning program (e.g., 1/2/3). BIL-04 sets per its policy. Recorded for audit.
Outstanding debt amount	The customer's outstanding debt at suspension time, supplied by BIL-04. Recorded for audit + downstream reporting (CVM, finance write-off analysis).
Auto-clear restrictions	Per SUB-LM-01 R-RES-6, all <code>active_restrictions[]</code> entries are cleared automatically when a subscription transitions to <code>SUSPENDED</code> (since suspension already revokes all service). This DD's BPMN includes the <code>sub-lm-put-active-restrictions</code> clear step.

1. What this is

The non-payment suspension operation. SUB-WF-SUSPEND-NP-01 is one of the 10 operation kinds in the framework. Triggered exclusively by BIL-04 when a subscription crosses its dunning suspension threshold (grace period expired on an unpaid invoice, prepaid wallet drained past reactivation window, level-3 dunning escalation reached). System-driven only — no human or self-service trigger is accepted.

Outcomes: subscription status → `SUSPENDED`, all active restrictions cleared (per SUB-LM-01 R-RES-6), network access revoked (FUL-04 `SUSPEND_SERVICE`), suspension history row written with `originIntent = SUSPEND_NP + systemManaged = true`, customer notified by NOT-01, BIL-04 acknowledged via Kafka event.

What this DD owns

1. `subscription_suspend_np_config` **table** — per-operator config: customer-notification preference at suspension, debt-amount threshold reporting flags.
2. **BILLING_INTERNAL-only trigger gating** — only BIL-04's service account may call the trigger endpoint.
3. `originIntent = SUSPEND_NP + systemManaged = true` marker contract on the pause-history row (the row is owned by SUB-LM-01's pause-history table, but this DD defines the flags applied).
4. **Trigger REST endpoint** — `POST /api/subscriptions/{subscription_id}/suspend-np`.
5. **Domain Kafka events** — `SubscriptionSuspendedForNonPayment + SubscriptionSuspendNPRejected`.
6. **Suspend-NP workflow rules** — 11 rules in §4.

What this DD does NOT own

- **BPMN files** — owned by per-operator satellites.
- **Worker implementations** — catalogued in `DD_SUB-WF-FRAMEWORK-01-WORKERS`.
- **Subscription state machine + status transitions** — owned by SUB-LM-01 (states, transitions, restriction-clear behavior on suspend per R-RES-6).
- **Pause-history table itself** — owned by SUB-LM-01.
- **Dunning ladder orchestration** — owned by BIL-04. This DD is a callee.
- **Network suspension implementation** — owned by FUL-04 (`SUSPEND_SERVICE` intent).
- **Restriction additions before full suspension** — owned by SUB-WF-RESTRICT-01 (BIL-04 may add restrictions at L2 via that workflow before reaching L3 here).
- **Resume after debt settlement** — owned by SUB-WF-RESUME-01 (`DUNNING_RESUME` trigger).

Regional flexibility

Per-operator levers:

Lever	Examples for SUSPEND-NP
BPMN file	KE: standard 11-worker BPMN. Operators with simpler dunning policies may skip the <code>bil01-emit-intent</code> analytics call.
Drools package rules. <code>subscription.suspend-np.<operator></code>	KE: minimal rules (role-gating + dunning context validation).
<code>subscription_suspend_np_config</code> row	KE: customer notification enabled; debt-amount-warning threshold tier KES 5,000 for HIGH-tier flagging.
FOUNDATION_AUTH candidate groups	BILLING_INTERNAL resolves to operator-specific BIL-04 service accounts.

2. Tables overview

This DD owns one table: `subscription_suspend_np_config` (per-operator config).

Cross-operation audit anchor: `subscription_operation` (framework-owned).

Subscription master fields updated by this workflow (owned by SUB-LM-01): `status_code = SUSPENDED`, `current_transition_type = SUSPEND_NP` (during `PENDING_SUSPEND_NP` phase), `current_transition_reason_code = <dunningReasonCode>`, `last_status_changed_at`.

Pause-history row inserted (owned by SUB-LM-01 — same table SUB-WF-PAUSE-01 uses): `originIntent = SUSPEND_NP`, `systemManaged = true`, plus dunning context fields.

3. Schemas

3.1 subscription_suspend_np_config

One row per operator.

```
CREATE TABLE subscription_suspend_np_config (
  operator_code          TEXT PRIMARY KEY,
  customer_notification_enabled  BOOLEAN NOT NULL DEFAULT TRUE,
  debt_amount_warning_threshold NUMERIC(18,2) NULL,
  debt_amount_warning_currency TEXT NULL,
  updated_at            TIMESTAMPTZ NOT NULL DEFAULT NOW(),
  updated_by            TEXT NOT NULL,
  CHECK ((debt_amount_warning_threshold IS NULL) = (debt_amount_warning_currency IS NULL))
);
```

Column	Notes
<code>customer_notification_enabled</code>	When true, the workflow calls the <code>notification-send</code> worker with <code>notificationEventKind=SUBSCRIPTION_SUSPENDED_NON_PAYMENT</code> at suspension. When false, no notification is dispatched (operator handles communication out-of-band, e.g., via a separate batch from BIL-04). NOT-01 owns template, channel, language, and fallback chain resolution.
<code>debt_amount_warning_threshold</code>	When the outstanding debt at suspension exceeds this value, the <code>SubscriptionSuspendedForNonPayment</code> event payload sets <code>debtAmountTier = HIGH</code> (otherwise STANDARD). Used by CVM analytics + finance reporting to flag high-value churn risk. Null = no tiering.
<code>debt_amount_warning_currency</code>	Currency for the threshold; nullable iff threshold is nullable.

Sample data:

<code>operator_code</code>	<code>customer_notification_enabled</code>	<code>debt_amount_warning_threshold</code>	<code>debt_amount_warning_currency</code>
WIK	true	5000.00	KES
WUG	true	NULL	NULL
WTZ	true	NULL	NULL
YASSN	true	NULL	NULL

4. Rules

4.1 State precondition + transition

Rule	Description
R-SUB-WF-SUSPEND-NP-01-S-1	Suspension requires the subscription's <code>status_code = ACTIVE</code> . Other statuses (already <code>SUSPENDED</code> , <code>PENDING_*</code> , <code>TERMINATED</code> , <code>RETIRED</code> , <code>CREATED</code>) return 409 <code>INVALID_STATE_FOR_SUSPEND_NP</code> . Idempotent re-submission of the same idempotency key returns the existing <code>operation_id</code> .
R-SUB-WF-SUSPEND-NP-01-S-2	The transition is <code>ACTIVE → PENDING_SUSPEND_NP → SUSPENDED</code> . <code>PENDING_SUSPEND_NP</code> is a framework-internal transient state (per SUB-LM-01 catalog) used during the saga; downstream consumers should see only the commit to <code>SUSPENDED</code> .
R-SUB-WF-SUSPEND-NP-01-S-3	Per framework R-FW-1, this operation is mutually exclusive with other state-affecting operations on the same subscription. If another in-flight operation exists (e.g., a customer just triggered <code>PAUSE</code>), the workflow uses the framework's external-cancel-requested mechanism to cancel it before proceeding. <code>RESTRICT</code> operations are concurrent-safe per their own design and do not need cancellation.
R-SUB-WF-SUSPEND-NP-01-S-4	The subscription's <code>current_transition_type</code> is set to <code>SUSPEND_NP</code> during <code>PENDING_SUSPEND_NP</code> . <code>current_transition_reason_code</code> is set to the caller-supplied <code>dunningReasonCode</code> .

4.2 Trigger gating

Rule	Description
R-SUB-WF-SUSPEND-NP-01-T-1	The trigger endpoint is callable only by the <code>BILLING_INTERNAL</code> role. Any other actor (human or service) returns 403 <code>UNAUTHORIZED_TRIGGER</code> with the allowed role for diagnostic clarity.
R-SUB-WF-SUSPEND-NP-01-T-2	Request body MUST include <code>dunningReasonCode</code> (free-text from BIL-04's reason catalog), <code>dunningCycleReference</code> (BIL-04's internal correlation ID for the dunning wave), <code>outstandingDebtAmount</code> , <code>outstandingDebtCurrency</code> . Optional: <code>dunningEscalationLevel</code> (integer). Missing required fields → 400 <code>MISSING_REQUIRED_FIELD</code> .
R-SUB-WF-SUSPEND-NP-01-T-3	Idempotency: BIL-04 generates the idempotency key per dunning cycle. Re-submission of the same key returns the existing <code>operation_id</code> per framework R-FW-3.

4.3 Master fields + pause-history

Rule	Description
R-SUB-WF-SUSPEND-NP-01-M-1	On commit, the subscription's master fields are updated: <code>last_status_changed_at = NOW()</code> . <code>suspended_at</code> (if SUB-LM-01 has this field; otherwise inferable from <code>status_history</code>) is set via <code>sub-lm-put-master-fields</code> .
R-SUB-WF-SUSPEND-NP-01-M-2	A pause-history row is inserted into SUB-LM-01's pause-history table with <code>originIntent = SUSPEND_NP</code> , <code>systemManaged = true</code> , <code>dunningReasonCode</code> , <code>dunningCycleReference</code> , <code>dunningEscalationLevel</code> , <code>outstandingDebtAmount</code> , <code>outstandingDebtCurrency</code> , <code>suspendedAt = commitAt</code> . The row has no <code>cycleAnchorPolicy</code> (not applicable; <code>SUSPEND-NP</code> does not configure cycle behavior).
R-SUB-WF-SUSPEND-NP-01-M-3	All entries in <code>subscription.active_restrictions[]</code> are cleared at the same time (per SUB-LM-01 R-RES-6 — restrictions are auto-removed when transitioning to <code>SUSPENDED</code> since suspension already revokes all service). Done via <code>sub-lm-put-active-restrictions</code> with <code>restrictionsOperationKind = CLEAR_ALL</code> . The pre-clear array is captured in the Kafka event for audit.

4.4 Resume protection

Rule	Description
R-SUB-WF-SUSPEND-NP-01-R-1	A subscription with an open <code>SUSPEND_NP</code> -flagged pause-history row can only be resumed via SUB-WF-RESUME-01's <code>DUNNING_RESUME</code> trigger (BIL-04 after debt settlement) or <code>ADMIN_FORCE_RESUME</code> trigger (admin override). SUB-WF-RESUME-01's role gating enforces this; SUB-WF-SUSPEND-NP-01 only ensures the flags are correctly set on the pause-history row.

Rule	Description
R-SUB-WF-SUSPEND-NP-01-R-2	If an admin invokes ADMIN_FORCE_RESUME (exceptional case), the original SUSPEND_NP marker on the pause-history row is preserved (the row is closed, but the originIntent is not changed). This surfaces in audit + operations dashboards as "admin force-resumed a non-payment suspension" — a flag for compliance review.

5. REST API

5.1 Trigger non-payment suspension

POST /api/subscriptions/{subscription_id}/suspend-np HTTP/1.1

Authorization: Bearer <bil04-service-jwt>

Content-Type: application/json

Idempotency-Key: suspend-np-sub_01HZ_KAMAU-dunning-wave-2026-06-15-L3

```
{
  "operatorCode":      "WIK",
  "dunningReasonCode": "DUNNING_ESCALATION_T3",
  "dunningCycleReference": "bil04-dunning-wave-2026-06-15-L3",
  "dunningEscalationLevel": 3,
  "outstandingDebtAmount": 3500.00,
  "outstandingDebtCurrency": "KES",
  "notes": "Dunning L3 reached after L2 voice-bar did not produce payment within 14 days",
  "initiatingActorUserId": "bil04-svc-account",
  "initiatingActorRole": "BILLING_INTERNAL",
  "initiatingWorkflowKind": "BIL_04_DUNNING",
  "initiatingWorkflowRef": "bil04-dunning-wave-2026-06-15-L3"
}
```

→ 202 Accepted

```
{
  "operationId":      "subop_01HZ_KAMAU_SUSPNP",
  "operatorCode":      "WIK",
  "subscriptionId":    "sub_01HZ_KAMAU_FIBER",
  "operationKind":      "SUSPEND_NP",
  "bpmnProcessKey":     "sub-suspend-np-wik",
  "bpmnProcessInstanceId": "pi_01HZ_KAMAU_SUSPNP",
  "currentState":      "VALIDATING",
  "startedAt":          "2026-06-15T02:00:00+03:00"
}
```

Unauthorized trigger:

→ 403 Forbidden

```
{
  "code":      "UNAUTHORIZED_TRIGGER",
  "allowedRoles": ["BILLING_INTERNAL"],
  "actorRole":  "CALL_CENTER"
}
```

Invalid state:

→ 409 Conflict

```
{
  "code":      "INVALID_STATE_FOR_SUSPEND_NP",
  "currentStatus": "SUSPENDED",
  "operationId":  "subop_01HZ_KAMAU_SUSPNP_PRIOR"
}
```

6. Kafka events emitted

Event	Topic	Fires on	Consumers
SubscriptionSuspendedForNonPayment	sophix.subscription.subscription.suspendednonpayment	Successful commit	NOT-01 (customer notification per operator config), BIL-04 (acknowledges its dunning step), CVM analytics (churn-risk modeling), EM-04 reporting, finance reporting (debt-tier flagging)
SubscriptionSuspendNPRejected	sophix.subscription.subscription.suspendednonpaymentrejected	Workflow failure (validation, billing, fulfillment)	BIL-04 (retry or escalate), ICN-01 ops alert, EM-04

Sample SubscriptionSuspendedForNonPayment:

```
{
  "eventType": "SubscriptionSuspendedForNonPayment",
  "aggregateId": "sub_01HZ_KAMAU_FIBER",
  "timestamp": "2026-06-15T02:00:18+03:00",
  "payload": {
    "operatorCode": "WIK",
    "subscriptionId": "sub_01HZ_KAMAU_FIBER",
    "customerId": "cust_01HZ_KAMAU",
    "operationId": "subop_01HZ_KAMAU_SUSPNP",
    "priorStatus": "ACTIVE",
    "newStatus": "SUSPENDED",
    "suspendedAt": "2026-06-15T02:00:18+03:00",
    "dunningReasonCode": "DUNNING_ESCALATION_T3",
    "dunningCycleReference": "bil04-dunning-wave-2026-06-15-L3",
    "dunningEscalationLevel": 3,
    "outstandingDebtAmount": 3500.00,
    "outstandingDebtCurrency": "KES",
    "debtAmountTier": "STANDARD",
    "clearedRestrictions": ["OUTGOING_VOICE_BARRED"],
    "pauseHistoryRowId": "pause_01HZ_KAMAU_SUSPNP",
    "originIntent": "SUSPEND_NP",
    "systemManaged": true
  }
}
```

Sample SubscriptionSuspendNPRejected:

```
{
  "eventType": "SubscriptionSuspendNPRejected",
  "aggregateId": "sub_01HZ_KAMAU_FIBER",
  "payload": {
    "operatorCode": "WIK",
    "subscriptionId": "sub_01HZ_KAMAU_FIBER",
    "operationId": "subop_01HZ_KAMAU_SUSPNP_FAIL",
    "rejectionReasonCode": "FULFILLMENT_FAILED",
    "rejectionReasonDetail": "FUL-04 returned FAILED on Internet service suspend (Huawei NCE timeout)",
    "subscriptionRevertedTo": "ACTIVE",
    "dunningCycleReference": "bil04-dunning-wave-2026-06-15-L3"
  }
}
```

7. Cross-DD coordination

7.1 Upstream caller (single)

Caller	Endpoint	Trigger value	Role
BIL-04 Dunning	POST .../suspend-np	(caller carries dunningReasonCode etc., no separate "trigger" enum here — this DD has only one trigger pattern, system-driven from dunning)	BILLING_INTERNAL

7.2 Sibling modules

Module	Direction	Contract
SUB-WF-FRAMEWORK-01 (parent)	This DD composes framework primitives	subscription_operation row created; BPMN started
SUB-WF-FRAMEWORK-01-WORKERS	This DD's BPMNs reference workers by topic	Workers used: drools-decide, sub-lm-cancel-in-flight-operation (if any), sub-lm-put-pending-status, sub-lm-compute-cycle-window, bil01-emit-intent, ful-call-restrict (intent=SUSPEND_SERVICE), sub-lm-put-active-restrictions (clear all), sub-lm-put-master-fields, sub-lm-commit-state, kafka-emit, notification-send, subscription-operation-finalize
SUB-LM-01	Workers write status + master fields + pause-history + clear restrictions	Status flip, master fields update, pause-history insert with originIntent=SUSPEND_NP, restrictions clear

Module	Direction	Contract
BIL-01	bil01-emit-intent worker	SUSPEND_NP_INTENT — billing records suspension state, freezes cycle accrual per operator policy, may emit analytics BillableEvent (no charge)
BIL-04 Dunning	Calls trigger endpoint + consumes events	Drives at L3 escalation; consumes SubscriptionSuspendedForNonPayment to acknowledge step
FUL-04	ful-call-restrict worker with intent=SUSPEND_SERVICE	Network-level suspension (AAA disable, may rate-zero OLT)
NOT-01	notification-send worker	Customer notification per operator config
SUB-WF-RESUME-01	Sibling DD — handles eventual resume after debt settlement	Reads pause-history row's originIntent=SUSPEND_NP to gate resume triggers
CVM / EM-04	Consume Kafka events	Churn analytics + debt-tier reporting

7.3 Build order

This DD deploys after:

1. SUB-WF-FRAMEWORK-01 (parent + WORKERS) v1.0
2. SUB-LM-01 v1.0
3. FUL-04 v1.0 (SUSPEND_SERVICE intent)
4. BIL-01 (SUSPEND_NP_INTENT support)
5. At least one FLOW satellite (FLOW-WIK v1.0)

7.4 Stub debt + open coordination items

- **WUG / WTZ / YASN FLOW satellites** — pending operator deep dives.
- **BIL-04 Dunning ladder L3 integration** — BIL-04 calls this DD's trigger endpoint. Captured in BIL-04 v1.4.
- **BIL-CFG-01 SUSPEND_NP_INTENT BillableEvent definitions** — pending BIL-CFG-01 v1.0 promotion.
- **SUB-WF-RESUME-01 DUNNING_RESUME contract** — pending DD; this DD's pause-history flags must be read correctly by SUB-WF-RESUME-01 to enforce resume gating.

B — Current TZ

Status: To be filled in after codebase cartography review.

Workshop materials note that the legacy dunning flow drives suspension via AAA disable at the final dunning step. Whether the legacy codebase has a structured pause-history or distinguishes voluntary pause from dunning suspension requires source-code review. Until that cartography lands, this DD does not assert specific legacy structural claims.